

MAKKAH, Saudi Arabia

WMO: 410300

Lat: 21.437N

Lon: 39.770E

Elev: 787

StdP: 14.28

Time Zone: 3.00 (E03)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	62.7	64.7	37.7	34.2	81.2	41.2	39.2	84.1	15.0	82.1	12.4	81.5	2.4	0	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	113.4	76.2	111.5	76.0	109.7	76.0	84.5	101.7	83.2	100.5	81.9	99.5	4.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	163.7	95.9	78.9	154.3	95.1	77.2	145.5	94.3	49.7	101.0	48.0	100.3	46.6	99.8	93.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	10.9	9.1	DB	58.1	118.9	2.9	2.0	56.0	120.4	54.3	121.6	52.7	122.7	50.6	124.2	(4)
(5)			WB	49.6	86.3	2.0	2.1	48.1	87.8	46.9	89.0	45.8	90.2	44.3	91.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	89.4	77.1	79.6	83.1	89.4	95.4	98.0	97.8	97.7	97.0	92.1	85.1	80.0	(6)	
	DBStd	8.38	3.77	4.08	4.34	3.91	3.70	2.93	2.50	2.33	2.33	3.04	3.48	3.26	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	14386	841	830	1027	1182	1408	1439	1481	1480	1410	1305	1053	930	(10)	
	CDD65	8911	376	410	562	732	943	989	1016	1015	960	840	603	465	(11)	
	CDH74	129673	3059	3858	6461	10427	15248	16802	17392	16987	15517	12416	7173	4333	(12)	
(13)	CDH80	86719	1222	1787	3442	6532	10854	12486	12929	12523	11198	8061	3766	1919	(13)	
(14)	Wind	WSAvg	2.7	2.7	3.0	3.2	3.1	2.8	2.7	2.9	2.5	2.2	2.2	2.2	(14)	
(15) Precipitation	PrecAvg	4.2	0.6	0.1	0.2	0.4	0.3	0.0	0.1	0.3	0.2	0.5	0.9	0.5	(15)	
	PrecMax	9.1	2.2	1.6	1.3	3.5	1.7	0.1	0.8	2.2	1.5	2.8	6.9	2.1	(16)	
	PrecMin	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	1.7	0.7	0.3	0.3	0.7	0.4	0.0	0.2	0.6	0.4	0.8	1.5	0.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.6	97.2	102.5	107.7	116.2	116.7	114.9	113.4	114.4	111.0	101.8	95.4	(19)	
		MCWB	70.9	72.2	71.8	74.5	76.0	74.3	75.6	79.6	76.2	73.6	72.9	71.0	(20)	
	2%	DB	91.5	94.9	99.0	105.0	111.6	113.8	111.6	110.9	111.3	107.7	99.0	93.4	(21)	
		MCWB	70.8	72.3	71.8	74.2	75.3	75.5	75.9	78.4	76.4	74.0	73.8	71.7	(22)	
	5%	DB	89.5	92.9	96.9	102.8	109.3	111.4	109.7	109.1	109.2	105.6	97.2	91.6	(23)	
		MCWB	70.4	71.7	71.8	74.0	75.0	75.5	75.4	77.9	77.3	74.6	73.6	71.8	(24)	
	10%	DB	87.3	89.9	94.7	100.5	107.0	109.3	107.8	107.3	107.2	102.6	95.2	89.7	(25)	
		MCWB	69.9	70.7	71.3	73.2	74.7	75.3	74.9	77.4	77.6	75.5	73.9	71.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.4	76.9	77.3	79.7	83.4	83.4	84.5	86.4	86.2	82.5	79.1	77.2	(27)	
		MCDB	86.0	92.4	93.6	99.9	105.4	105.7	102.2	104.6	103.7	99.2	92.8	87.6	(28)	
	2%	WB	73.8	75.1	75.6	78.0	80.3	81.1	82.4	84.5	84.4	80.6	77.7	75.6	(29)	
		MCDB	84.8	89.3	92.1	98.0	102.0	104.5	100.6	101.3	101.3	96.3	91.0	86.1	(30)	
	5%	WB	72.8	73.8	74.3	76.7	78.7	79.4	80.5	83.3	82.8	79.3	76.7	74.4	(31)	
		MCDB	83.7	87.1	90.6	96.2	100.8	103.7	99.8	99.6	99.8	94.7	89.9	85.3	(32)	
	10%	WB	71.7	72.7	73.1	75.5	77.2	77.9	78.8	82.0	81.5	78.4	75.7	73.4	(33)	
		MCDB	82.0	85.1	89.1	95.0	100.1	102.6	99.7	98.4	98.6	93.7	88.7	84.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	21.1	23.0	23.5	24.2	24.7	22.3	21.1	22.9	23.4	20.5	19.6	(35)	
		MCDBR	22.3	23.5	25.0	25.1	26.6	26.8	24.2	23.0	25.2	26.1	22.6	21.8	(36)	
	5% WB	MCWBR	9.5	9.6	8.8	8.9	9.3	9.4	8.9	9.0	8.6	7.7	7.1	8.8	(37)	
		MCWBR	19.3	21.1	21.7	22.4	23.2	23.7	20.3	19.9	21.3	22.7	20.2	19.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.439	0.483	0.509	0.542	0.589	0.608	0.639	0.664	0.618	0.545	0.478	0.449	(40)	
		taud	2.129	1.990	1.914	1.862	1.745	1.697	1.673	1.632	1.698	1.850	2.023	2.095	(41)	
	Ebn at Noon		260	256	255	248	235	229	222	217	224	235	246	251	(42)	
		Edh at Noon	45	55	62	66	74	77	79	82	76	63	50	46	(43)	
(44) All-Sky Solar Radiation	RadAvg		1458	1741	2064	2250	2386	2474	2355	2208	2081	1873	1547	1393	(44)	
(45)	RadStd		77	75	75	85	76	54	56	49	60	55	67	60	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+387	+389
(47)	Regional (2 neighbors)	+0.86	+1.26	-2.23	N/A	N/A	N/A	N/A	N/A	+338	+310

Nomenclature: See separate page